



Koneru Lakshmaiah Education Foundation

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: REAL TIME OPERATING SYSTEMS – (24IN3103HF)

A.Y. 2026 - 2027

TITLE OF THE PROJECT	A REAL-TIME PRIORITY SCHEDULING FOR AUTONOMOUS WILDFIRE DETECTION & RESPONSE FRAMWWORK	
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Abstract:

Wildfires spread rapidly and unpredictably, and delayed detection or poorly coordinated emergency response often results in preventable loss of life and property. Existing wildfire monitoring systems typically rely on periodic sensor polling without any formal mechanism to guarantee that a critical fire alert is processed within a bounded time, meaning high priority alerts can be silently delayed behind lower priority tasks under real sensor load.

This work proposes BLAZE-RT, a real-time task scheduling framework that models routine sensor monitoring as periodic low priority tasks and fire-detection events as high priority tasks with strict response deadlines, applying Rate-Monotonic scheduling and Earliest-Deadline-First scheduling algorithms to guarantee timely response, while explicitly modeling and resolving priority inversion, a scenario where a lower priority task blocks a higher priority task, through a technique called priority inheritance.

The system is implemented in the Python programming language with a simulated fire-spread grid based on cellular automaton modeling, combined with a physical sensor layer that streams live

temperature and flame readings into the scheduler, alongside a live visual dashboard showing task execution and deadline compliance. Experiments are conducted under varying sensor-load conditions, comparing scheduled execution against a naive first-come-first-served baseline. The framework is evaluated using the deadline-miss rate, response latency, and system utilization, and results demonstrate that formal real-time scheduling significantly reduces missed critical alerts compared to the naive baseline, confirming that deadline-aware scheduling meaningfully improves response reliability under load. The proposed framework offers a practical, low-cost, and reproducible blueprint for deadline-guaranteed emergency response systems in resource-constrained edge computing environments, with clear extensibility to other safety-critical monitoring domains beyond wildfire detection.